Dogs with Minimum Scores

In a dog beauty contest with ‘n’ participants, each dog is scored based on ‘m’ different criteria. Each criterion has a maximum possible score. A dog is automatically disqualified from the overall contest if it does not meet the minimum score requirement for any one of the criteria – it will not be evaluated for that criterion, but it may still be evaluated for others.

Write a program that identifies the dogs which achieved exactly the minimum score for any of the criteria (even if they meet this threshold for more than one criterion, they should only appear once in the list).

Input

The first line contains ‘n’ (number of dogs, 1 ≤ n ≤ 100) and ‘m’ (number of criteria, 1 ≤ m ≤ 100). The second line contains ‘m’ integers, representing the maximum scores for each criterion (1 ≤ maxpoint[i] ≤ 100). The third line contains ‘m’ integers, representing the minimum score required for each criterion (1 ≤ minpoint[i] ≤ maxpoint[i]).

The next ‘n’ lines contain ‘m’ integers each, where each number represents the score of a dog for a specific criterion (0 ≤ points[i,j] ≤ maxpoint[j]).

Output

The number of dogs passing with the minimum score should be printed, then print the list of the indexes in the order they appeared (separated by space).

Example

Input	*Output*
6 8 9 9 9 9 9 9 9 9 5 5 5 5 5 5 5 5 8 4 6 6 6 6 6 6 7 5 7 6 6 6 6 5 6 6 6 5 5 5 5 6 8 6 8 7 7 7 7 6 8 9 6 6 6 6 6 6 8 6 6 6 6 6 6 1	2 2 3